Project Dept-Heater Sales Yearly Wise (2024-2025)

1. Project Overview and Objective

This project focuses on cleaning, transforming, and analyzing raw datasets using Microsoft Excel, followed by the development of an interactive dashboard in Power BI to generate actionable business insights.

The primary objective is to demonstrate effective data pre-processing techniques, including handling missing values, removing duplicates, standardizing formats, and structuring data for analysis. Excel is used for initial data wrangling and transformation, ensuring data quality and consistency.

Subsequently, the refined dataset is imported into Power BI, where interactive visualizations and dashboards are created. These dashboards enable users to explore key performance indicators (KPIs), identify trends, and support data-driven decision-making.

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2. Data Sources

- **Source Description and Timeline:** Project Dept-Heater Sales Yearly Wise (2024-2025) Real Time Data
- **Domain**: Sales

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3. Problem Statement

- To analyze the yearly sales performance of heaters for the period 2024–2025, the organization needs a concise summary of sales trends, key variations, and overall performance insights
- The goal is to understand how heater sales have changed over the year, identify any significant increases or decreases, and highlight important patterns that can support decision-making for production, inventory management, and future sales planning

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4. Attribute (Column /Features) Details:

Column Name	Attribute Description	Data Type	Data Category
Sales Office	Internal code of the sales office	String	Categorical
Branch Name	Branch location name	String	Categorical
Proper Case Material Description	Customer / Supplier name	String	Categorical
External Material Group	Material classification (e.g., STORAGE)	String	Categorical
Quantity	Number of units billed	Decimal / Numeric	Numerical
Year	Billing year	Integer	Numerical
Billing Date	Actual billing date	Date	Date

Billing Document Number	Unique billing invoice number	Integer / String (if leading zeros)	Categorical / Identifier
Netvalue	Net price (Before GST)	Decimal	Numerical
Price Per Unit	Unit price	Decimal	Numerical
Total Amount Spend Per Heater	Cost per heater	Decimal	Numerical
Price Status	Price type (e.g., Standard Price)	String	Categorical
Material Code	Internal material code	Integer	Identifier / Categorical
Material	Product model (e.g., 15GH-R)	String	Categorical
Description	Heater color description	String	Categorical
Sales Order Number	Sales order reference	Integer / String	Identifier / Categorical
Customer Code	Unique customer ID	Integer / String	Identifier / Categorical
Total Amount (Included GST)	Final billing amount including GST	Decimal	Numerical

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5.ExploreTools

Excel: Data Explore

Data Explore	
Total Number of Sales	966
Total Amount of Sales Including Gst	229008990.8
Total Amount of Sales Net Price	194053964.2
Average Amount of Sales Including Gst	237069.3487
Average Amount of Sales Net Price	200884.0209
Average Quantity of Heater Sales	43
Mode of Sales Product Code Wise	501300
Group Wise Customer Count	0.972049689
Total Amount from Delhi Branch	80192751.15
Total Amount from Chennai Branch	33323039.57
Total Amount from Mumbai Branch	39625189.01
Number of Purchase from Delhi Branch	156

Number of Purchase from Mumbai Branch	154
Total Amount From Model Wise 25GH-L	40654676.37
Total Amount From Model Wise 25GH-R	43861561.96
Number of Delhi Branch who Purchased storage Heater	156
Number of Mumbai Branch who Purchased storage Heater	147
Most Popular Product Category	STORAGE
Branch Wise Most Customers	Chennai Branch
Year Wise Sales(2025)	126608290.7
Year Wise Sales(2024)	102400700.2
Company Spend Amount Year Wise(2025)	83447011.69
Company Spend Amount Year Wise(2024)	67483849.32

1. Total Number of Sales – 966

"Our dataset contains **966 successful sales transactions**, giving us a strong sample size to understand customer behaviour and branch performance."

=COUNTA('Cleaned Data Sheet'!A2:A967)

2. Total Sales Amount Including GST – ₹22.90 Cr

"The company generated **₹22.9 Crores** in revenue, inclusive of GST. This highlights the total billing strength achieved across all branches."

=SUM('Cleaned Data Sheet'!R2:R967)

3. Total Net Sales (Without GST) – ₹19.40 Cr

"The net revenue — after removing GST — stands at **₹19.40 Crores**, representing the true business earnings."

=SUM('Cleaned Data Sheet'!I2:I967)

4. Average Sales Amount (Including GST) – ₹2,37,069

"On average, each customer spends around **₹2.37 Lakhs**, showing a high-value purchase pattern."

=AVERAGE('Cleaned Data Sheet'!R2:R967)

5. Average Sales Amount (Net Price) – ₹2,00,884

"Excluding GST, each sale brings in about **₹2 Lakhs**, confirming strong pricing and product value."

=AVERAGE('Cleaned Data Sheet'!I2:I967)

6. Average Quantity of Heater Sales – 43 Units

"Customers typically purchase around **43 heaters per order**, indicating bulk purchase behaviour."

=INT(AVERAGE('Cleaned Data Sheet'!E2:E967))

7. Most Sold Product Code – 501300

"The product with code **501300** is the **top-selling item**, reflecting customer preference and high market demand."

=MODE('Cleaned Data Sheet'!M2:M967)

8. Group Wise Customer Count – 97%

"Nearly **97%** of all customers belong to a particular key customer group, showing a focused customer segment."

=COUNTIF('Cleaned Data Sheet'!D2:D967,"STORAGE")/E4

9. Branch-Wise Revenue Contribution

- **Delhi Branch:** ₹8.01 Cr
- **Chennai Branch:** ₹3.33 Cr
- **Mumbai Branch:** ₹3.96 Cr

"Delhi leads the revenue chart with the highest contribution."

Example Formula=SUMIF('Cleaned Data Sheet'!B2#,"Delhi Branch",'Cleaned Data Sheet'!R2:R967)

10. Model-Wise Sales Amount

- **25GH-L:** ₹4.06 Cr
- **25GH-R:** ₹4.38 Cr
  "The model **25GH-R** slightly outperforms **25GH-L** in overall revenue."

Example=SUMIF('Cleaned Data Sheet'!N2:N967,"25GH-L",'Cleaned Data Sheet'!R2:R967)

11. Branch-wise Storage Heater Purchases

- **Delhi:** 156
- **Mumbai:** 147

"Delhi purchases the most **Storage Heater products**."

12. Most Popular Product Category – STORAGE

“The Storage category is the most demanded across all customers.”

Example=COUNTIFS('Cleaned Data Sheet'!D2:D967,"STORAGE",'Cleaned Data Sheet'!B2#,"Delhi Branch")

13. Branch With Most Customers – Chennai

“Chennai has the **largest number of unique customers**, even if total sales amount is lower.”

=INDEX('Cleaned Data Sheet'!D2:D967,MODE(MATCH('Cleaned Data Sheet'!D2:D967,'Cleaned Data Sheet'!D2:D967,0)))

14. Year-wise Sales Performance

- **2025:** ₹12.66 Cr
- **2024:** ₹10.24 Cr

“The business grew by more than **₹2.4 Cr** from FY 2024 to FY 2025.”

=SUMIF('Cleaned Data Sheet'!F2#,"2025",'Cleaned Data Sheet'!R2:R967)

15. Company Spend (Cost) Year-wise

- **2025:** ₹8.34 Cr
- **2024:** ₹6.74 Cr

“Company expenses increased due to higher sales and operational activities.”

=SUMIF('Cleaned Data Sheet'!F2#,"2025",'Cleaned Data Sheet'!K2:K967)

6. Data Pre-Processing (Excel)

- **Excel:** Fact Table

Sales Office	City	Branch Name
B001	Chennai	CMO Chennai
B100	Chennai	Chennai Branch
B101	Madurai	Madurai Branch
B102	Coimbatore	Coimbatore Branch
B103	Ernakulam - Kerala	Ernakulam Branch
B104	Bangalore	Bangalore Branch
B105	TELANGANA	SECUNDERABAD Branch
B106	New Delhi	Delhi Branch
B107	Ghaziabad	Ghaziabad Branch
B108	Chandigarh	Chandigarh Branch

B109	Kolkatta	Kolkatta Branch

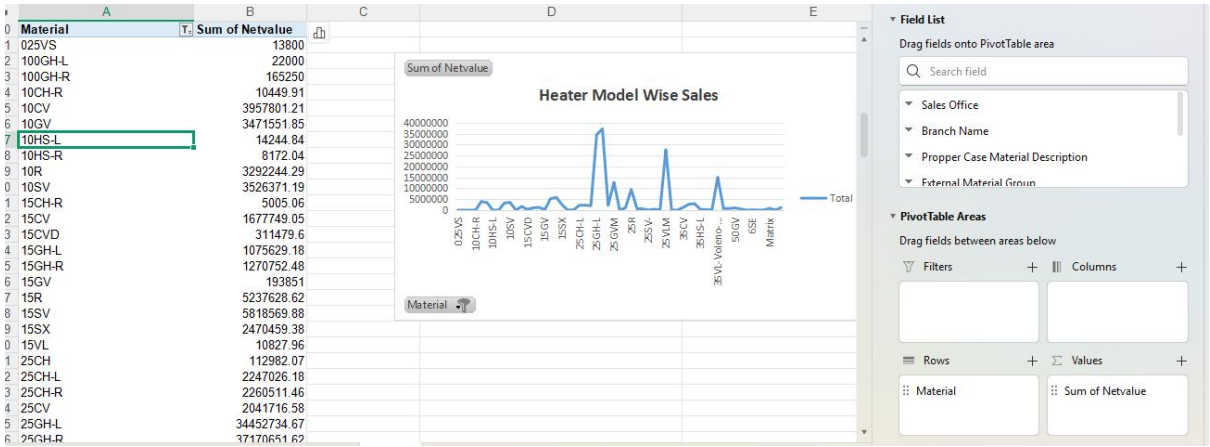
6. Data Pre-Processing (Excel)

- **Excel:** Dimensional Table

Material Code	Material	Price Per Unit
500119	25GV	3,895
500121	50GV	3,895
500122	70GV	3,895
500127	15GV	3,895
500128	25GV	3,895
500130	50GV	3,895
500138	35GH-R	3,895
500141	100GH-R	3,895

6. Data Pre-Processing (Excel)

Pivot Table&Pivot Chart



5. Data Pre-Processing (Excel)

Vlookup Using Excel

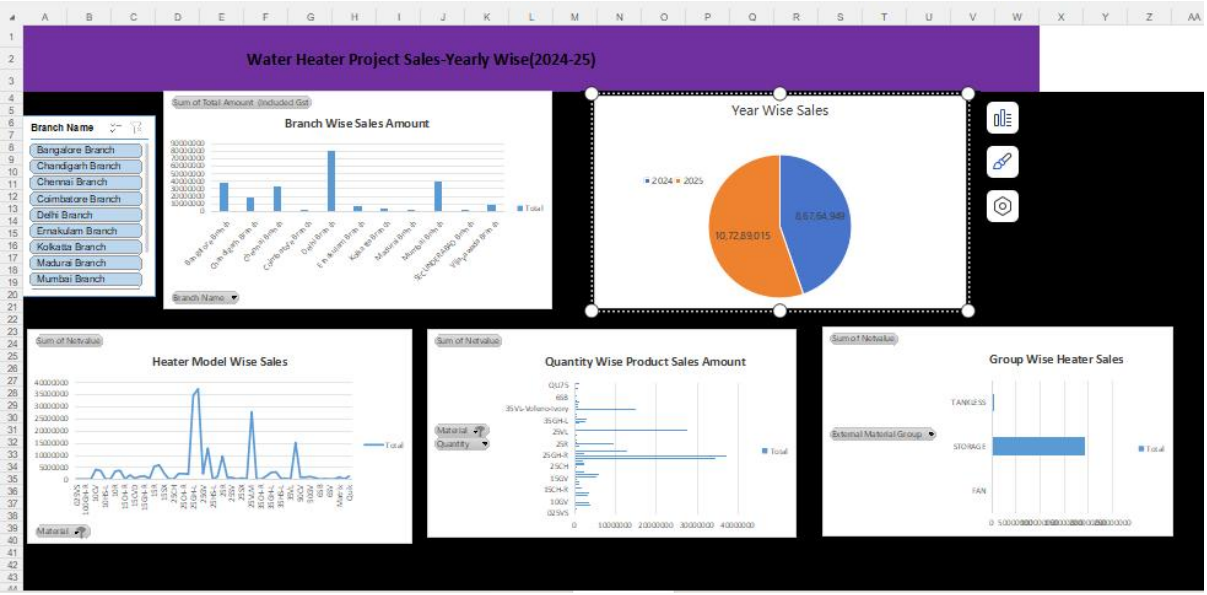
B2 | =VLOOKUP(A2:A967,'Branch Name'!1:1048576,3,1)

	A	B	C	D	E	Ye
1	Sales Office	Branch Name	Propper Case Material Description	External Material Group	Quantity	
2	B106	Delhi Branch	Sobha Limited	STORAGE	1.000	
3	B103	Ernakulam Branch	Sobha Limited	STORAGE	1.000	
4	B111	Mumbai Branch	Cosmic Distri. Pvt Ltd - Bulk	STORAGE	1.000	
5	B111	Mumbai Branch	Cosmic Distri. Pvt Ltd - Bulk	STORAGE	1.000	
6	B104	Bangalore Branch	Sobha Limited	STORAGE	1.000	
7	B104	Bangalore Branch	Sobha Limited	STORAGE	1.000	
8	B104	Bangalore Branch	Sobha Limited	STORAGE	1.000	
9	B104	Bangalore Branch	Sobha Limited	STORAGE	1.000	
10	B104	Bangalore Branch	Sobha Limited	STORAGE	1.000	
11	B104	Bangalore Branch	Sobha Limited	STORAGE	1.000	
12	B104	Bangalore Branch	Sobha Limited	STORAGE	1.000	
13	B106	Delhi Branch	Sobha Limited	STORAGE	1.000	
14	B106	Delhi Branch	Sobha Limited	STORAGE	1.000	
15	B106	Delhi Branch	Sobha Limited	STORAGE	1.000	
16	B111	Mumbai Branch	Cosmic Distri. Pvt Ltd - Bulk	STORAGE	1.000	
17	B104	Bangalore Branch	Prestige Estates Projects Limited	STORAGE	1.000	
18	B111	Mumbai Branch	Cosmic Distri. Pvt Ltd - Bulk	TANKLESS	1.000	
19	B111	Mumbai Branch	Cosmic Distri. Pvt Ltd - Bulk	STORAGE	1.000	
20	B111	Mumbai Branch	Swayam Realtors & Traders Llp	STORAGE	1.000	
21	B111	Mumbai Branch	Cosmic Distri. Pvt Ltd - Bulk	STORAGE	1.000	
22	B111	Mumbai Branch	Cosmic Distri. Pvt Ltd - Bulk	STORAGE	1.000	
23	B111	Mumbai Branch	Cosmic Distri. Pvt Ltd - Bulk	STORAGE	1.000	
24	B111	Mumbai Branch	Cosmic Distri. Pvt Ltd - Bulk	STORAGE	1.000	
25	B111	Mumbai Branch	Cosmic Distri. Pvt Ltd - Bulk	STORAGE	1.000	

Formula=VLOOKUP(A2:A967,'Branch Name'!1:1048576,3,1)

6. ExploreTols

This dashboard provides a comprehensive overview of the **Water Heater Project sales performance for the year 2024–25**, presenting key insights through multiple charts and slicers for interactive analysis.



7. Revenue

Both 2024 and 2025 show a profitable trend.The increase in sales and controlled spending from 2024 to 2025 led to a significant improvement in total profit.This indicates strong business growth, better cost optimization, and an upward financial trajectory.

Year Wise Sales(2025)	126608290.7
Year Wise Sales(2024)	102400700.2

Company Spend Amount Year Wise(2025)	83447011.69
Company Spend Amount Year Wise(2024)	67483849.32

Company Spend Amount Year Wise	Profit Status
4,31,61,279	Profit
3,49,16,851	Profit

- **Power BI:** Data modelling, DAX calculations, visualization, and interactive dashboard creation.

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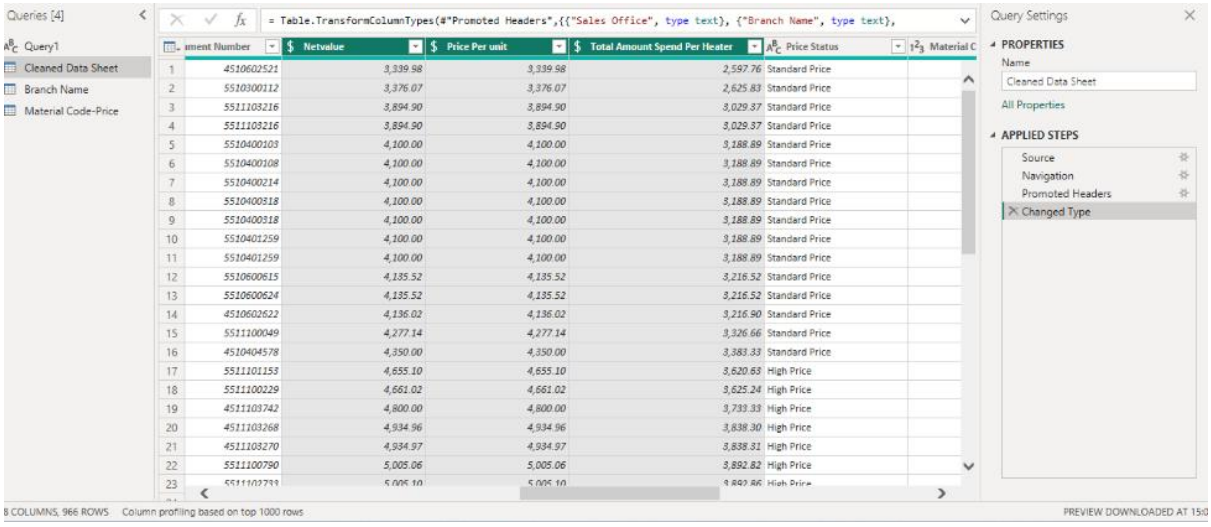
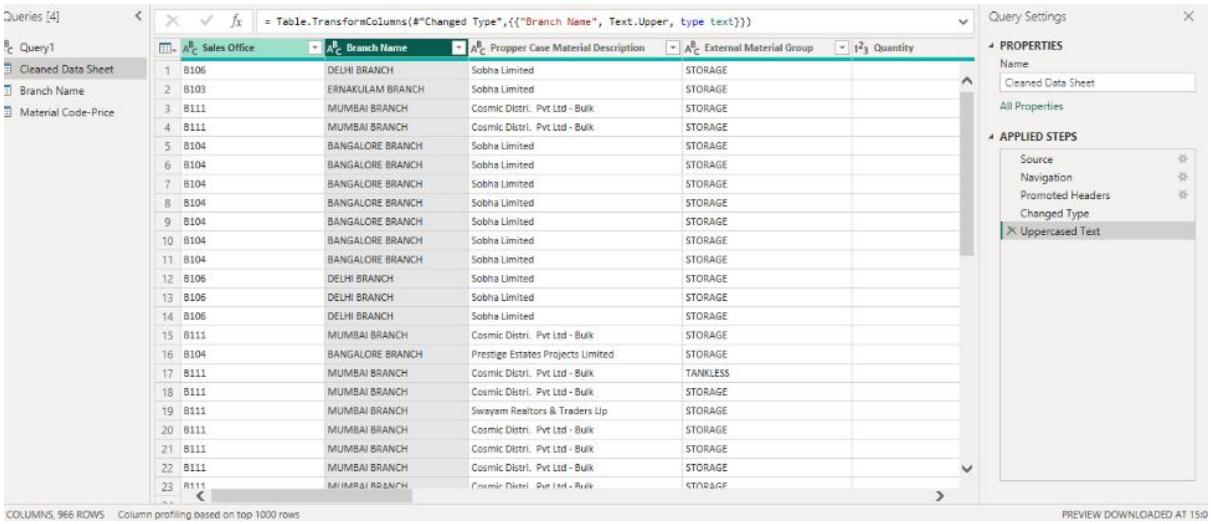
8. Data Modelling and DAX (Power BI)

Fixed Decimal Number in Power BI is a data type used to store numbers with exact precision, up to 4 decimal places, without rounding errors.It is mainly used for financial, currency, and monetary calculations where accuracy is critical.

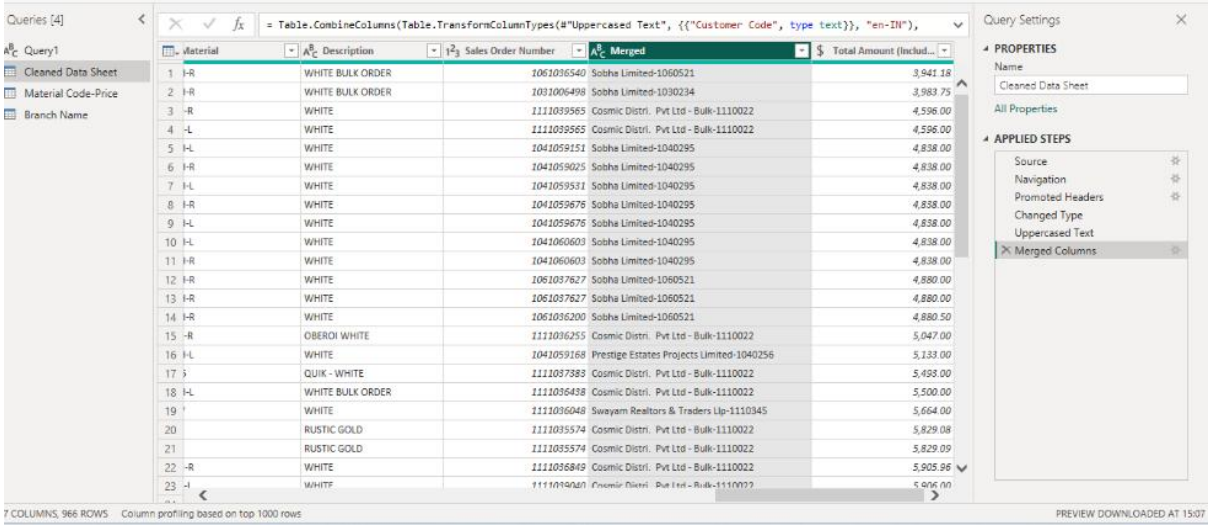


= Table.TransformColumnTypes(
#"Promoted Headers",
{
{"Sales Office", type text},
{"Branch Name", type text}
}
)

The "Branch Name" column has been formatted into UPPER CASE to maintain consistent capitalization across all records. This transformation ensures uniform data presentation, eliminates variations caused by mixed or improper casing, and improves data cleanliness for reporting and analysis. By converting all branch names to uppercase, it becomes easier to group, filter, and compare values accurately within Power BI.



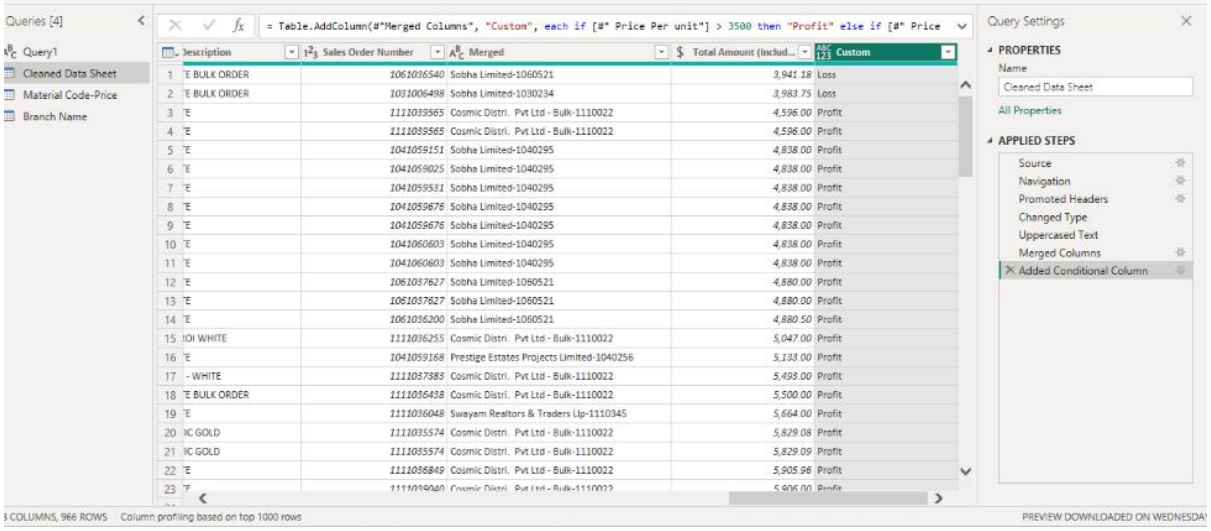
The Location column is created by merging the Sales Office and Branch Name columns to provide a combined identifier for each record. This helps simplify reporting and improves readability by showing both details in a single field.



A new conditional column named **"Profit Status"** is created to classify each record based on the value in the **"Profit"** column.
The logic applied is as follows:

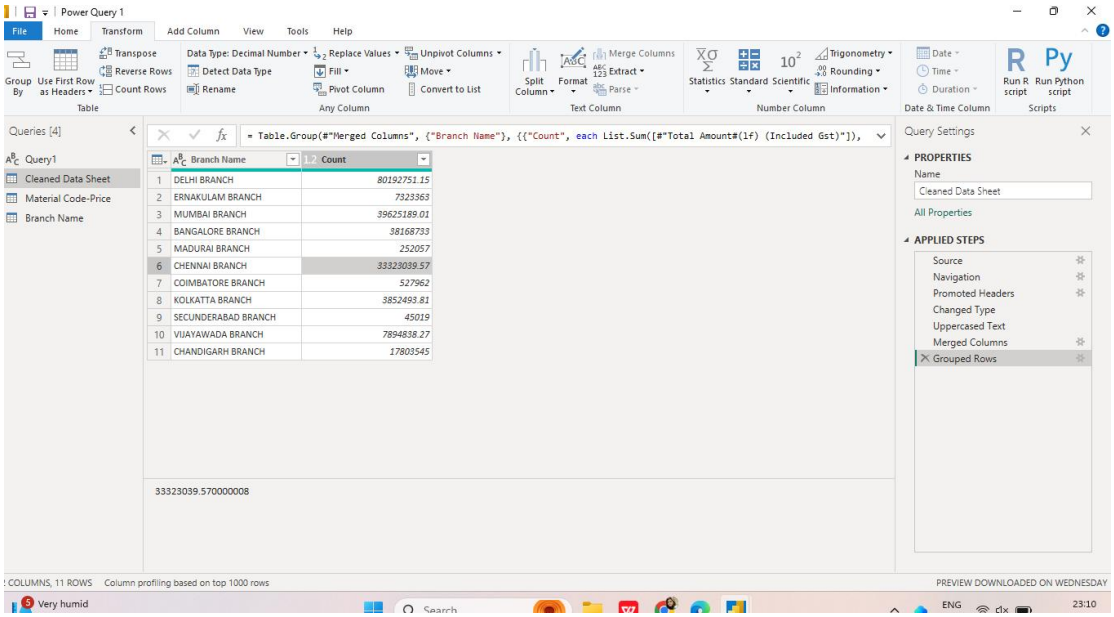
- If the **Profit** value is **less than 3500**, the **Profit Status** will be labeled as **"Loss"**.
- If the **Profit** value is **greater than 3500**, the **Profit Status** will be labeled as **"Profit"**.

This conditional logic helps in quickly identifying whether a transaction or product performance falls under **profit-making** or **loss-making** categories for further analysis.



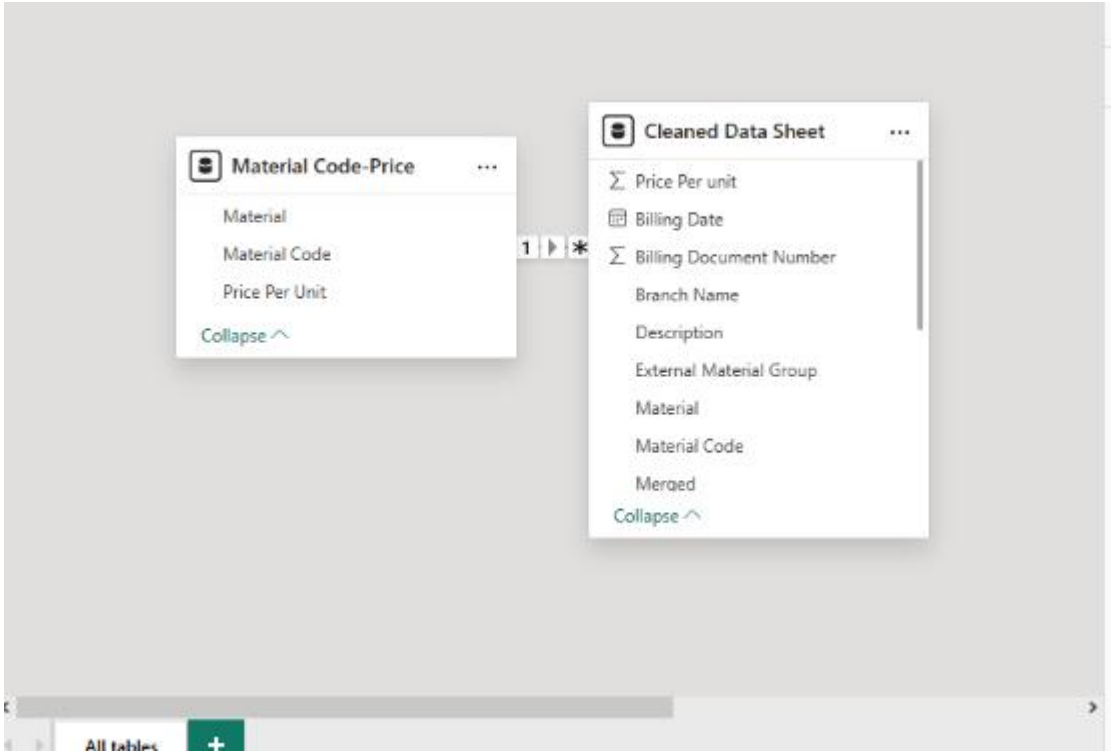
Grouping and Aggregating Data:

This step involves summarizing the dataset by Branch Name to understand the overall sales performance of each branch. The data is grouped based on the Branch Name column, and two key aggregations are performed:



● Build a relationship between the “Cleaned Data sheet”and Material Code Price tables based on the ‘Material’ column and “Material Code” Click "Manage relationships" and ensure this relationship is active.

To ensure accurate data modeling and seamless calculations across tables, a relationship was created between the Cleaned Data sheet and the Material Code Price table. The relationship was established using the common fields — ‘Material’ from the Cleaned Data sheet and ‘Material Code’ from the Material Code Price table.



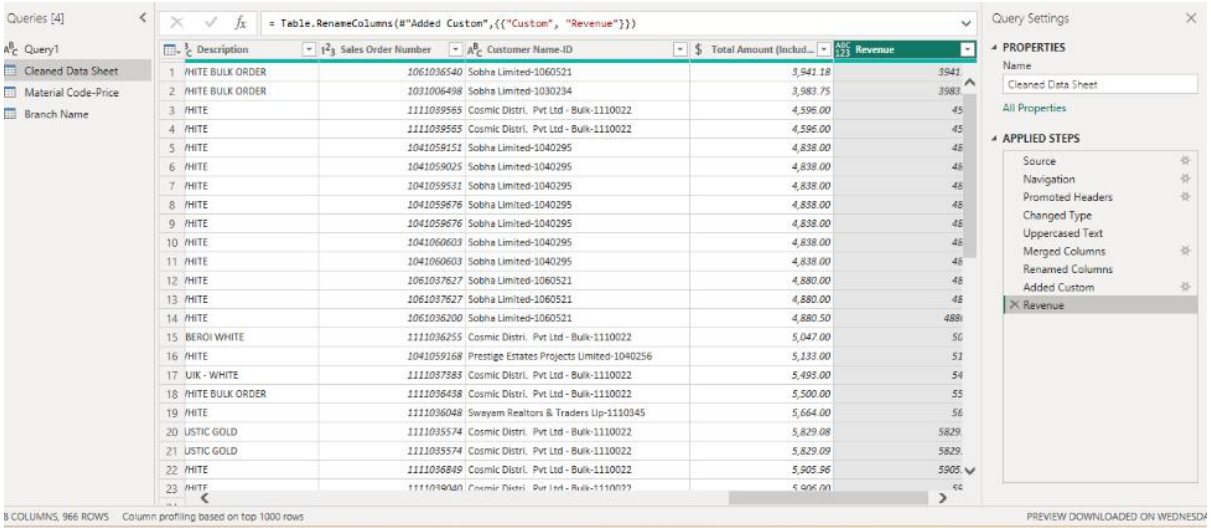
Calculated Column find Revenue

● Create a Calculated Column for Total Amount Spend Company vs Total Amount Sales

Purpose of the Column

The calculated column is used to:

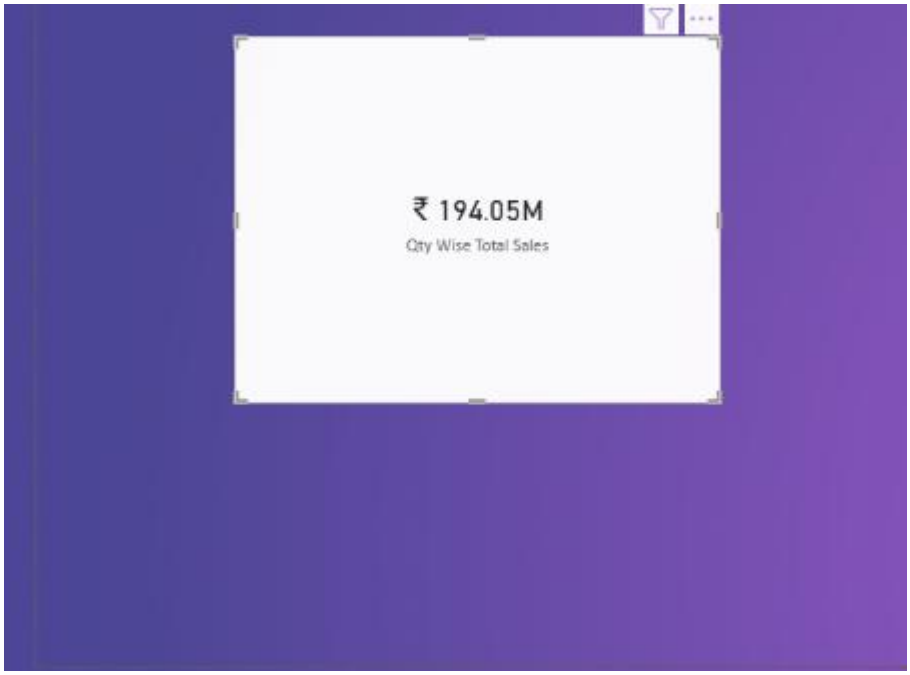
- Measure the **actual revenue generated** per transaction or per record.
- Understand how much money the company **earns after subtracting expenses/spend**.
- Support further analysis such as profit status, trend analysis, and financial performance.



9. Analysis and Visualizations (Power BI)

Visualization Line:

● Visualization Line: Define a Branch Wise Sales



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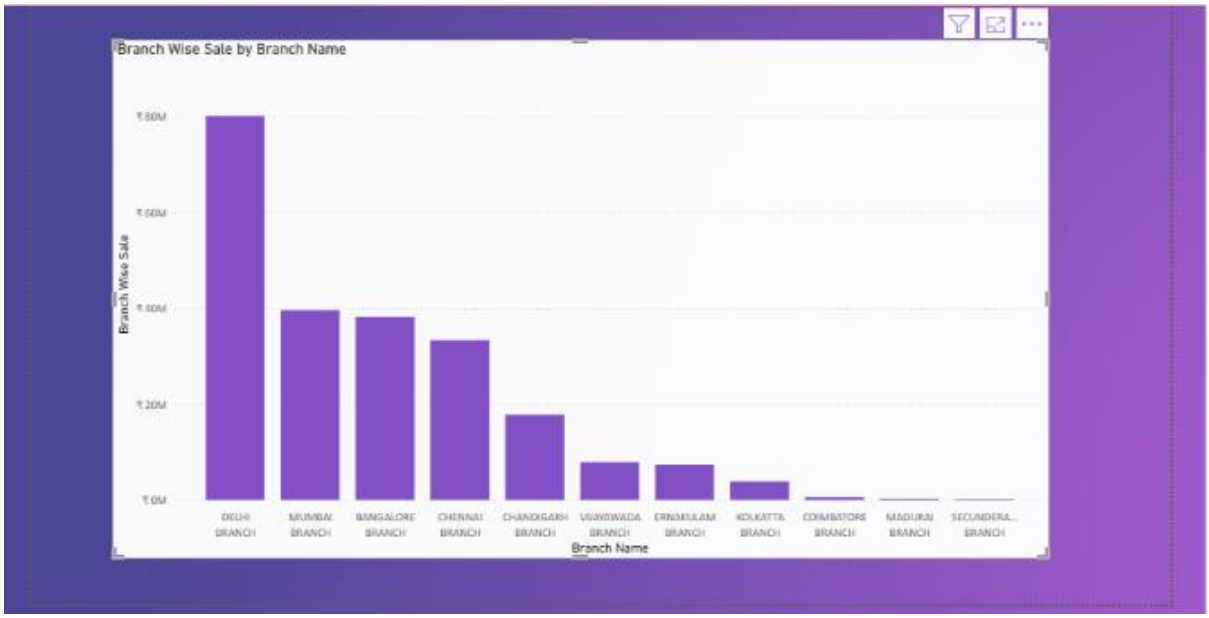
Visualization Line:

● Visualization Line: Define a Branch Wise Sales

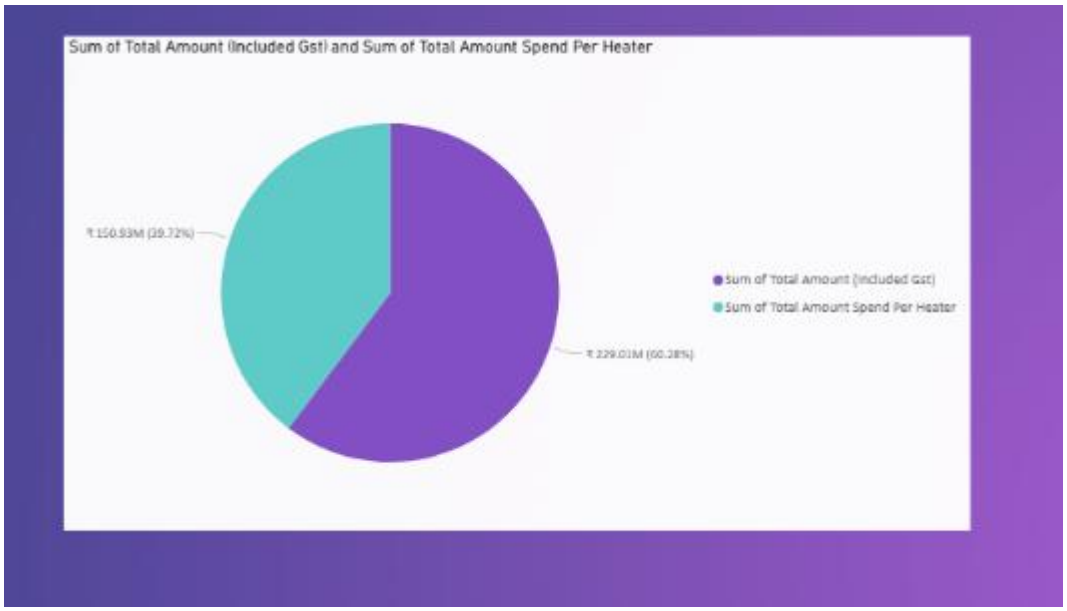
Stacked Column Chart:

● Stacked Column Chart:Branch Wise Sum of Total Amount Sales



Pie Chart:

●The chart highlights that **customer billing including GST** forms the majority of the total amount.
Meanwhile, **company spending per heater** forms a smaller but significant share.
This comparison helps understand **profit margins, cost distribution**, and **sales performance**.

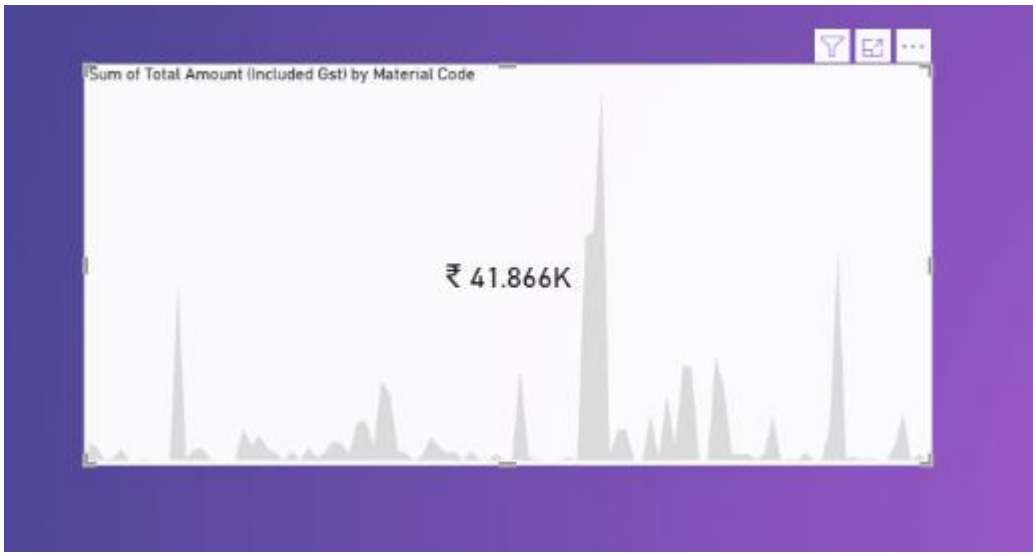


KPI Chart:

This chart displays the **Sum of Total Amount (Included GST) by Material Code**.
It appears as an **area chart**, where each spike represents the **total sales amount (including GST)** for different material codes.

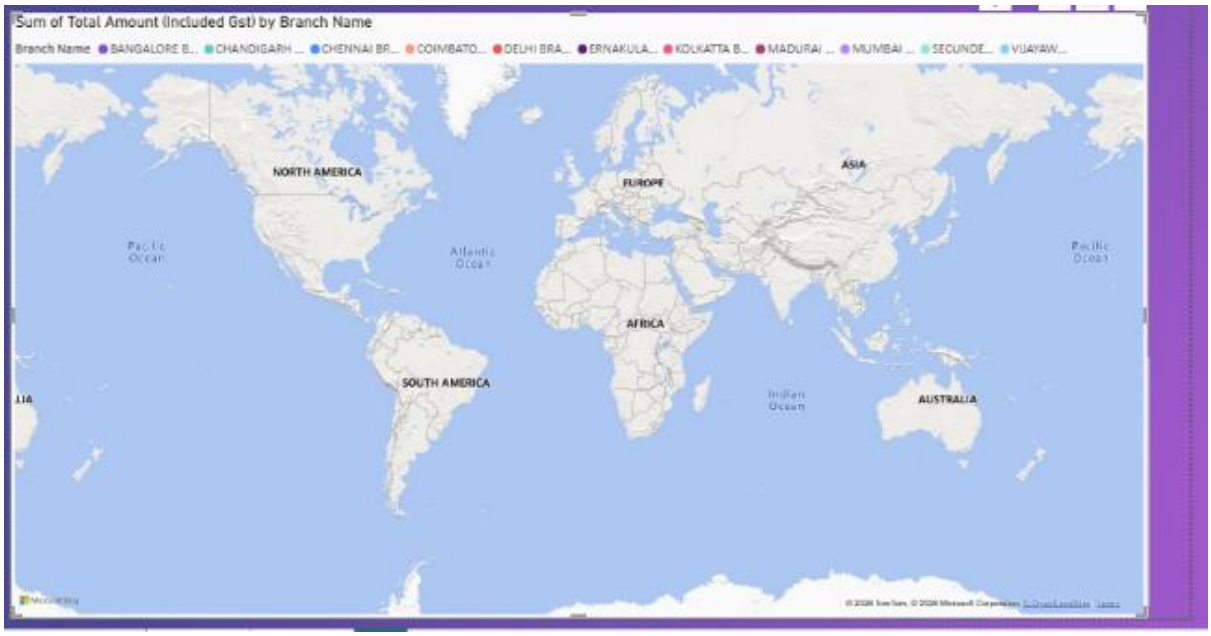
- The large spikes indicate **material codes with very high sales amounts**.
- Smaller spikes represent **material codes with lower sales amounts**.
- In the center of the visual, the value **₹ 41.866K** is shown — this represents the **total or highlighted sum of sales including GST**.

Overall, the visualization helps quickly identify which material codes contribute the most to the total sales amount and highlights variations in sales across different materials.



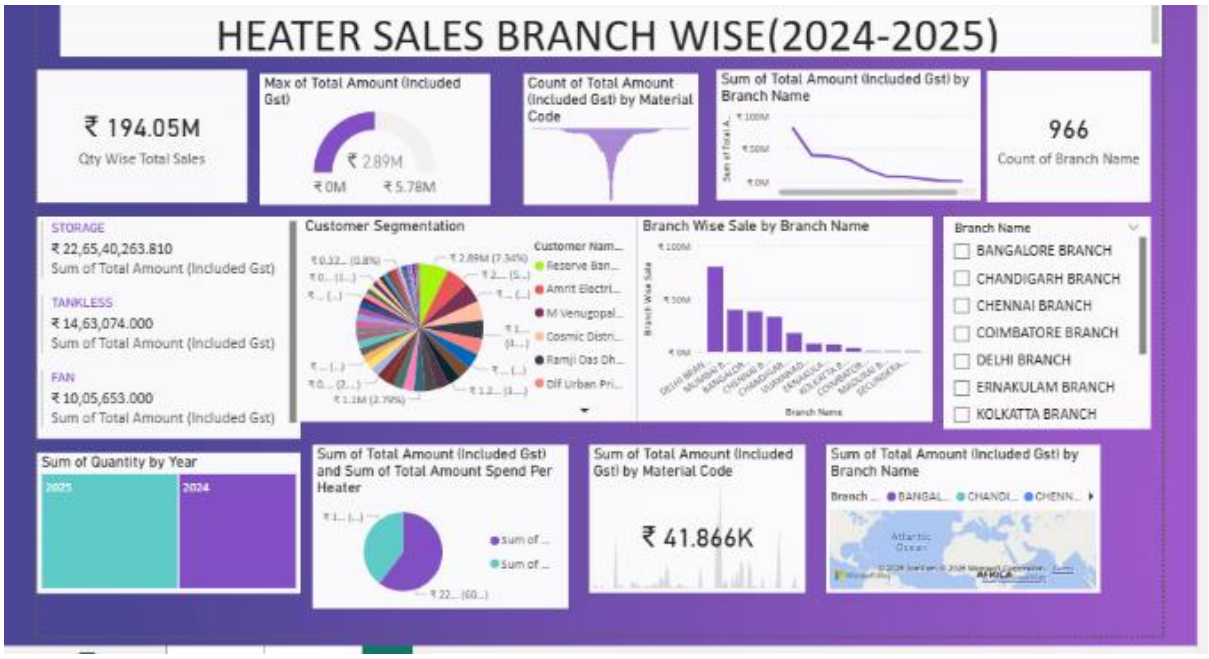
MapChart:

●This map visual shows the total amount of sales (including GST) distributed across different branch locations. Each branch is represented by a colored marker plotted on the world map.



Visualization :

This dashboard provides a comprehensive analysis of heater sales across multiple branches for the years 2024 and 2025, combining key performance indicators, segmentation charts, branch-wise comparisons, and geographic insights. Each visual helps users quickly understand sales behavior, revenue distribution, customer patterns, and material-level performance.



9. Summarization

Summarizes the actual heater sales data such as total sales amount, branch-wise sales, year-wise sales comparison, customer segmentation, and material-wise performance.Explains the reasons behind sales differences across branches and years, identifying which materials, customer groups, or locations caused high or low performance.Uses past sales trends to forecast future performance, predicting which branches, materials, and customer segments are likely to grow or decline.Suggests actions to improve future sales—such as focusing on high-performing branches, optimizing stock for top materials, targeting key customer segments, and improving weaker branch operations.

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10. Conclusions

The **Heater Sales Branch Wise (2024–2025)** analysis highlights strong overall performance with total sales reaching **₹194.05M**. The sales distribution indicates that a few key branches contribute significantly to revenue, with **Bangalore, Chennai, and Delhi** showing the highest sales volumes. Customer segmentation reveals a diverse customer base, but a major share of the total amount is driven by **high-value customers**, which is crucial for revenue stability.

Material-wise sales show that a limited group of material codes contributes to the highest GST-included amounts, suggesting strong product demand for specific heater models. The **year-wise quantity comparison** indicates steady demand from 2024 to 2025, confirming consistent market performance.

The dashboard also highlights balanced spending patterns, with spend per customer and branch-wise sales aligning well with overall revenue growth. Geographical distribution shows strong market presence across major metropolitan branches.

Overall, the analysis confirms that the company has a **healthy and growing heater sales performance**, driven by top-performing branches, key product lines, and high-value customers. Strategic focus on expanding lower-performing branches and optimizing product mix can further enhance revenue in the coming years.